Node.Js Pactical Assignment - 3

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1) Develop a Shopping cart in PHP with following requirements:

Admin Site:

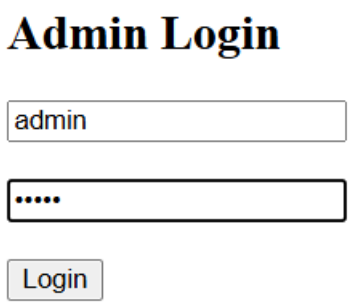
Login:

Code: (login.php)

```
<?php
session_start();
require "../config/db.php";
$message = "";
if ($_SERVER['REQUEST_METHOD'] == "POST") {
   $username = $_POST['username'];
   $password = $_POST['password'];
   $stmt = $conn->prepare(
      "SELECT *
       FROM admins
       WHERE username = ?"
   );
   $stmt->bind_param("s", $username);
   $stmt->execute();
   $admin = $stmt->get_result()->fetch_assoc();
   if (
      $admin &&
      password_verify(
         $password,
         $admin['password']
      )
   ) {
      $_SESSION['admin_id'] =
         $admin['id'];
```

```
        header("Location: dashboard.php");
        exit;
    } else {
        $message = "Invalid login.";
    }
}
?>
<h2>Admin Login</h2
<p><?= htmlspecialchars($message) ?></p>
<form method="post">
    <input name="email" type="email" placeholder="Email" required>
    <br><br>
    <input name="password" type="password" placeholder="Password" required>
    <br><br>
    <button>
        Login
    </button>
</form>
```

Output :



2) Manage Category (CRUD)

Categories.php

```
<?php
require "config/auth.php";
admin_required();
require "config/db.php";
$message = "";
$editCategory = null;
if (isset($_POST["add_category"])) {
   $name = trim($_POST["name"]);
   if ($name == "") {
      $message = "Category name is required.";
   } else {
      $stmt = $conn->prepare(
         "INSERT INTO categories(name)
          VALUES(?)"
      );
      $stmt->bind_param("s", $name);
      if ($stmt->execute()) {
         $message = "Category added successfully.";
      } else {
         $message = "Error: " . $stmt->error;
      }
   }
}
if (isset($_POST["update_category"])) {
   $id = intval($_POST["id"]);
   $name = trim($_POST["name"]);
   if ($name == "") {
      $message = "Category name is required.";
   } else {
      $stmt = $conn->prepare(
         "UPDATE categories
          SET name = ?
          WHERE id = ?"
      );
      $stmt->bind_param("si", $name, $id);
      if ($stmt->execute()) {
         $message = "Category updated successfully.";
      } else {
         $message = "Error: " . $stmt->error;
      }
   }
```

```
}
if (isset($_GET["delete"])) {
    $id = intval($_GET["delete"]);
    $stmt = $conn->prepare(
        "DELETE FROM categories
         WHERE id = ?"
    );
    $stmt->bind_param("i", $id);
    if ($stmt->execute()) {
        $message = "Category deleted successfully.";
    } else {
        $message = "Error: " . $stmt->error;
    }
}
if (isset($_GET["edit"])) {
    $id = intval($_GET["edit"]);
    $stmt = $conn->prepare(
        "SELECT id, name
         FROM categories
         WHERE id = ?"
    );
    $stmt->bind_param("i", $id);
    $stmt->execute();
    $result = $stmt->get_result();
    if ($result->num_rows === 1) {

        $editCategory = $result->fetch_assoc();
    }
}
$result = $conn->query(
    "SELECT id, name
     FROM categories
     ORDER BY id DESC"
);
?>
<!DOCTYPE html>
<html>
<head>
    <title>Manage Categories</title>
</head>
<body>
    <h1>Manage Categories</h1>
    <a href="dashboard.php">
        ← Dashboard
```

```
</a>
<br><br>
<?php if ($message != ""): ?>
  <p style="color:green;">
    <?= htmlspecialchars($message) ?>
  </p>
<?php endif; ?>
<?php if ($editCategory): ?>
  <h2>Edit Category</h2>
  <form method="POST">
    <input type="hidden" name="id" value="<?= $editCategory["id"] ?>">
    <input type="text" name="name" value="<?= htmlspecialchars($editCategory["name"]) ?>" required>
    <button type="submit" name="update_category">
      Update
    </button>
    <a href="categories.php">
      Cancel
    </a>
  </form>
<?php else: ?>
  <h2>Add Category</h2>
  <form method="POST">
    <input type="text" name="name" placeholder="Category name" required>
    <button type="submit" name="add_category">
      Add Category
    </button>
  </form>
<?php endif; ?>
<hr>
<h2>Category List</h2>
<table border="1" cellpadding="10">
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Action</th>
  </tr>
  <?php while ($category = $result->fetch_assoc()): ?>

    <tr>
      <td>
        <?= $category["id"] ?>
      </td>

      <td>
```

```
            <?= htmlspecialchars($category["name"]) ?>
          </td>
          <td>
            <a href="categories.php?edit=<?= $category["id"] ?>">
              Edit
            </a>
            |
            <a href="categories.php?delete=<?= $category["id"] ?>"
              onclick="return confirm('Delete this category?')">
              Delete
            </a>

          </td>
        </tr>
      <?php endwhile; ?>
    </table>
</body>
</html>
```

Output :

Manage Categories

← Dashboard

Add Category

Category name | Add Category

Category List

ID	Name	Action
3	Laptops	Edit | Delete
2	Mobiles	Edit | Delete
1	Garments	Edit | Delete

3) Manage Products (Product data with pictures. Upload picture and resize them)

```
<?php
require "../config/auth.php";
admin_required();
require "../config/db.php";
$message = "";
if (isset($_POST["add_product"])) {
   $category_id = intval($_POST["category_id"]);
   $pname = trim($_POST["pname"]);
   $price = floatval($_POST["price"]);
   $picture = null;
   if (
      isset($_FILES["picture"]) &&
      $_FILES["picture"]["error"] === UPLOAD_ERR_OK
   ) {
      $tmpName = $_FILES["picture"]["tmp_name"];
      $imageInfo = getimagesize($tmpName);
      if ($imageInfo === false) {
         $message = "Uploaded file is not a valid image.";
      } else {
         $width = $imageInfo[0];
         $height = $imageInfo[1];
         $type = $imageInfo[2];
         if (
            $type != IMAGETYPE_JPEG &&
            $type != IMAGETYPE_PNG &&
            $type != IMAGETYPE_WEBP
         ) {
            $message = "Only JPG, PNG and WEBP images are allowed.";
         } else {
            if ($type == IMAGETYPE_JPEG) {
               $source = imagecreatefromjpeg($tmpName);
            } elseif ($type == IMAGETYPE_PNG) {
               $source = imagecreatefrompng($tmpName);
            } else {
               $source = imagecreatefromwebp($tmpName);

            }
            $maxSize = 500;
            $scale = min(
               $maxSize / $width,
               $maxSize / $height,
               1
            );
            $newWidth = intval($width * $scale);
            $newHeight = intval($height * $scale);
            $destination = imagecreatetruecolor(
               $newWidth,
               $newHeight
            );
```

```
            $white = imagecolorallocate(
                $destination,
                255,
                255,
                255
            );
            imagefill(
                $destination,
                0,
                0,
                $white
            );
            imagecopyresampled(
                $destination,
                $source,
                0,
                0,
                0,
                0,
                $newWidth,
                $newHeight,
                $width,
                $height
            );
            $uploadDir = __DIR__ . "/../uploads/products/";
            if (!is_dir($uploadDir)) {
                mkdir(
                    $uploadDir,
                    0777,
                    true
                );
            }
            $picture = uniqid("product_") . ".jpg";
            imagejpeg(
                $destination,
                $uploadDir . $picture,
                85
            );
            imagedestroy($source);
            imagedestroy($destination);
        }
    }
}
if ($message == "") {
    $stmt = $conn->prepare(
        "INSERT INTO products
        (category_id, pname, price, picture)
        VALUES (?, ?, ?, ?)"
    );
    $stmt->bind_param(
        "isds",
        $category_id,
        $pname,
        $price,
```

```
            $picture
        );
        if ($stmt->execute()) {
            $message = "Product added successfully.";
        } else {
            $message = "Error: " . $stmt->error;
        }
        $stmt->close();
    }
}

if (isset($_GET["delete"])) {
    $id = intval($_GET["delete"]);
    $stmt = $conn->prepare(
        "SELECT picture
         FROM products
         WHERE id = ?"
    );
    $stmt->bind_param(
        "i",
        $id
    );
    $stmt->execute();
    $result = $stmt->get_result();
    if ($result->num_rows === 1) {
        $product = $result->fetch_assoc();
     if (!empty($product["picture"])) {
            $file =
                __DIR__ .
                "/../uploads/products/" .
                $product["picture"];
            if (file_exists($file)) {
                unlink($file);
            }
        }
    }
    $stmt->close();
    $stmt = $conn->prepare(
        "DELETE FROM products
         WHERE id = ?"
    );
    $stmt->bind_param(
        "i",
        $id
    );
    if ($stmt->execute()) {
        $message = "Product deleted successfully.";
    } else {
        $message = "Error deleting product.";
    }
    $stmt->close();
}
$categories = $conn->query(
    "SELECT id, name
```

```
    FROM categories
    ORDER BY name"
);
$products = $conn->query(
   "SELECT
      products.id,
      products.pname,
      products.price,
      products.picture,
      categories.name AS category_name
    FROM products
    INNER JOIN categories
      ON products.category_id = categories.id
    ORDER BY products.id DESC"
);
?>
<!DOCTYPE html>
<html>
<head>
   <meta charset="UTF-8">
   <title>Manage Products</title>
</head>
<body>
<h1>Manage Products</h1>
<a href="../dashboard.php">
   ← Dashboard
</a>
<br><br>
<?php if ($message != ""): ?>
   <p style="color: green;">
      <?= htmlspecialchars($message) ?>
   </p>
<?php endif; ?>
<hr>
<h2>Add Product</h2>
<form
   method="POST"
   enctype="multipart/form-data"
>
   <label>
      Category
   </label>
   <br>
   <select
      name="category_id"
      required
   >
      <option value="">
         Select Category
      </option>
      <?php while ($category = $categories->fetch_assoc()): ?>
         <option
            value="<?= $category["id"] ?>"
         >
```

```
            <?= htmlspecialchars($category["name"]) ?>
          </option>
        <?php endwhile; ?>
      </select>
      <br><br>
      <label>
        Product Name
      </label>
      <br>
      <input
        type="text"
        name="pname"
        required
      >
      <br><br>

      <label>
        Price
      </label>
      <br>
      <input
        type="number"
        name="price"
        step="0.01"
        min="0"
        required
      >
      <br><br>
      <label>
        Product Picture
      </label>
      <br>
      <input
        type="file"
        name="picture"
        accept="image/jpeg,image/png,image/webp"
      >
      <br><br>
      <button
        type="submit"
        name="add_product"
      >

        Add Product

      </button>

    </form>
    <hr>
    <h2>Product List</h2>
    <table
      border="1"
      cellpadding="10"
      cellspacing="0"
```

```
>
  <tr>
    <th>
      ID
    </th>
    <th>
      Picture
    </th>
    <th>
      Product Name
    </th>
    <th>
      Category
    </th>
    <th>
      Price
    </th>
    <th>
      Action
    </th>
  </tr>
  <?php while ($product = $products->fetch_assoc()): ?>
    <tr>
      <td>
        <?= $product["id"] ?>
      </td>
      <td>
        <?php if (!empty($product["picture"])): ?>
          <img
            src="../uploads/products/<?= htmlspecialchars($product["picture"]) ?>"
            width="100"
            height="100"
            style="object-fit: contain;"
          >
        <?php else: ?>
          No Image
        <?php endif; ?>
      </td>
      <td>
        <?= htmlspecialchars($product["pname"]) ?>
      </td>
      <td>
        <?= htmlspecialchars($product["category_name"]) ?>
      </td>
      <td>
        ₹<?= number_format(
          $product["price"],
          2
        ) ?>
      </td>
      <td>
        <a
          href="products.php?delete=<?= $product["id"] ?>"
          onclick="return confirm('Are you sure you want to delete this product?')"
```

```
            >
               Delete
            </a>
          </td>
        </tr>
    <?php endwhile; ?>
</table>
</body>
</html>
```

OUTPUT

Manage Products

← Dashboard

Add Product

Category
Select Category

Product Name

Price

Product Picture
Choose File No file chosen

Add Product

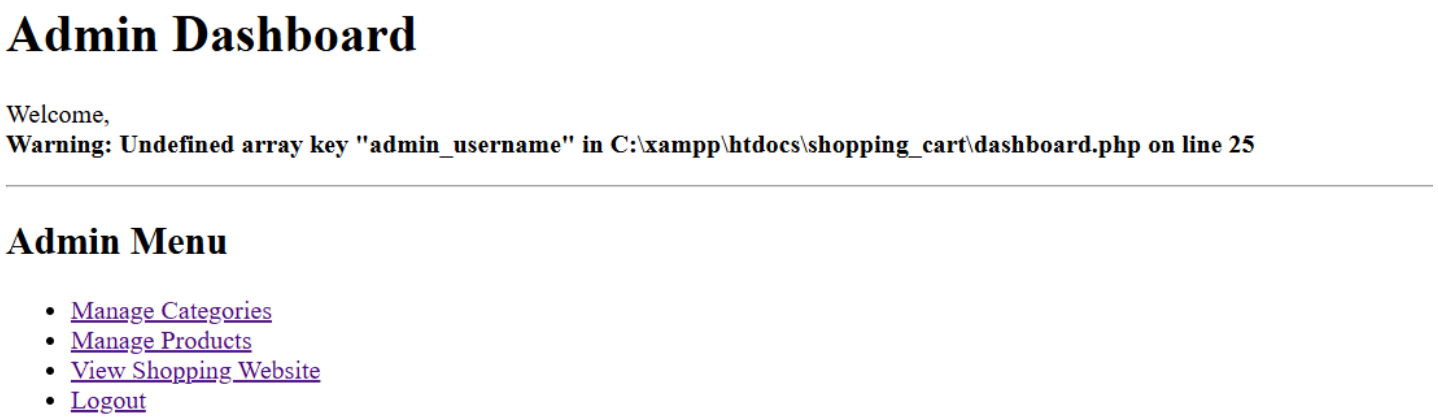
Product List

ID	Picture	Product Name	Category	Price	Action
1		dfnsr	Garments	₹1,234.00	Delete

4) Logout

```
<?php
session_start();
session_unset();
session_destroy();
header("Location: login.php");
exit;
?>
```

Output:



User Site:

1) List Categories

```
<?php
session_start();
require "config/db.php";
$category_id = isset($_GET["id"])
   ? intval($_GET["id"])
   : 0;
if ($category_id <= 0) {
   die("Invalid category ID.");
}
$stmt = $conn->prepare(
   "SELECT id, name
    FROM categories
```

```
    WHERE id = ?"
);
$stmt->bind_param("i", $category_id);
$stmt->execute();
$result = $stmt->get_result();
if ($result->num_rows == 0) {
   die("Category not found.");
}
$category = $result->fetch_assoc();
$category_name = $category["name"];
?>
<div class="container">
   <h2>
      <?= htmlspecialchars($category_name) ?>
   </h2>
   <div id="products">
      Loading products...
   </div>
</div>
<script>
function loadProducts(page = 1) {
   fetch(
      "ajax/products.php?category_id=<?= $category_id ?>&page=" + page
   )
   .then(response => {
      if (!response.ok) {
         throw new Error("HTTP error: " + response.status);
      }
      return response.text();
   })
   .then(data => {
      document.getElementById("products").innerHTML = data;
   })
   .catch(error => {
      console.error(error);
      document.getElementById("products").innerHTML =
         "<p>Error loading products.</p>";
   });
}
function changePage(page) {
   loadProducts(page);
}
```

```
loadProducts(1);
</script>
<?php
?>
```

Output:

Garments



dfnsr

₹1,234.00

Category: Garments

3) Register user (Use CAPTCHA)

```
<?php
session_start();
require "config/db.php";

if (!isset($_SESSION["num1"]) || !isset($_SESSION["num2"])) {
    $_SESSION["num1"] = rand(1, 9);
    $_SESSION["num2"] = rand(1, 9);
}
$message = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = trim($_POST["name"] ?? "");
    $email = trim($_POST["email"] ?? "");
    $password = $_POST["password"] ?? "";
    $captcha = intval($_POST["captcha"] ?? 0);
    $correct_captcha =
```

```
    $_SESSION["num1"] + $_SESSION["num2"];
 if ($captcha != $correct_captcha) {
    $message = "Incorrect CAPTCHA.";
 }
 elseif ($name == "" || $email == "" || $password == "") {
    $message = "All fields are required.";
 }
 elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $message = "Please enter a valid email.";
 }
 else {
    $check = $conn->prepare(
       "SELECT id FROM users WHERE email = ?"
    );
    $check->bind_param("s", $email);
    $check->execute();
    $check_result = $check->get_result();
    if ($check_result->num_rows > 0) {
       $message = "Email already registered.";
    }
    else {
       $hashed_password =
          password_hash(
             $password,
             PASSWORD_DEFAULT
          );
       $stmt = $conn->prepare(
          "INSERT INTO users
          (name, email, password)
          VALUES (?, ?, ?)"
       );
       $stmt->bind_param(
          "sss",
          $name,
          $email,
          $hashed_password
       );
       if ($stmt->execute()) {
          $message =
             "Registration successful. You can now login.";

          $_SESSION["num1"] = rand(1, 9);
```

```
            $_SESSION["num2"] = rand(1, 9);
        }
        else {
            $message =
                "Registration failed: " . $stmt->error;
        }
    }
  }
}
?>
<div class="container">
  <h2>User Registration</h2>
  <?php if ($message != ""): ?>
    <p>
      <?= htmlspecialchars($message) ?>
    </p>
  <?php endif; ?>
  <form method="POST">
    <label>Name</label>
    <br>
    <input
      type="text"
      name="name"
      required
    >
    <br><br>
    <label>Email</label>
    <br>
    <input
      type="email"
      name="email"
      required
    >
    <br><br>
    <label>Password</label>
    <br>
    <input
      type="password"
      name="password"
      required
    >
    <br><br
```

```
    <label>
      CAPTCHA:
      <?= $_SESSION["num1"] ?>
      +
      <?= $_SESSION["num2"] ?>
      =
    </label>
    <br>
    <input
      type="number"
      name="captcha"
      required
    >
    <button type="submit">
      Register
    </button>
  </form>
  <p>
    Already have an account?
    <a href="login.php">
      Login
    </a>
  </p>
</div>
```

Output:



4) Login

```
<?php
session_start();
require "config/db.php";
$message = "";
if ($_SERVER["REQUEST_METHOD"] == "POST") {
   $email = trim($_POST["email"] ?? "");
   $password = $_POST["password"] ?? "";
   if ($email == "" || $password == "") {
      $message = "Email and password are required.";
   }
   else {
      $stmt = $conn->prepare(
         "SELECT id, name, email, password
          FROM users
          WHERE email = ?"
      );
      $stmt->bind_param("s", $email);
      $stmt->execute();
      $result = $stmt->get_result();
      if ($result->num_rows == 1) {
         $user = $result->fetch_assoc();
         if (
            password_verify(
               $password,
               $user["password"]
            )
         ) {

            $_SESSION["user_id"] =
               $user["id"];
            $_SESSION["user_name"] =
               $user["name"];
            $_SESSION["user_email"] =
               $user["email"];
            header("Location: index.php");
            exit;
         }
         else {
```

```
            $message =
               "Invalid email or password.";
         }
      }

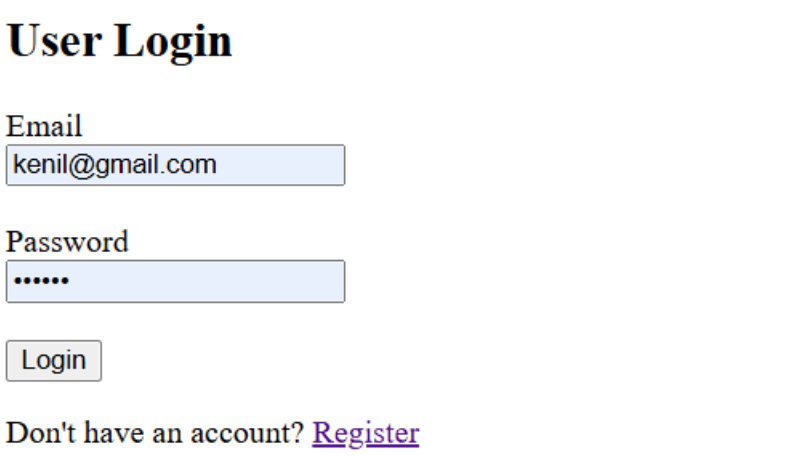
      else {
         $message =
            "Invalid email or password.";
      }
   }
}

?>

<div class="container">
   <h2>User Login</h2>
   <?php if ($message != ""): ?>
      <p>
         <?= htmlspecialchars($message) ?>
      </p>
   <?php endif; ?>
   <form method="POST">
      <label>Email</label>
      <br>
      <input
         type="email"
         name="email"
         required
      >
      <br><br>
      <label>Password</label>
      <br>
      <input
         type="password"
         name="password"
         required
      >
      <br><br>
      <button type="submit">
         Login
      </button>
   </form>
```

```
  <p>
    Don't have an account?
    <a href="register.php">
      Register
    </a>
  </p>
</div>
<?php
?>
```

Output:



2) Read data from students.xml file and store them in students table of MySQL.

students.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<students>

  <student>
    <roll_no>101</roll_no>
    <name>Rahul</name>
    <email>rahul@gmail.com</email>
    <course>BCA</course>
    <age>21</age>
  </student>

  <student>
```

```
    <roll_no>102</roll_no>
    <name>Priya</name>
    <email>priya@gmail.com</email>
    <course>MSc ICT</course>
    <age>22</age>
  </student>

  <student>
    <roll_no>103</roll_no>
    <name>Amit</name>
    <email>amit@gmail.com</email>
    <course>BCA</course>
    <age>20</age>
  </student>
</students>
```

xmp_import.php

```
<?php
require "config/db.php";
$file = "xml/students.xml";
if (!file_exists($file)) {
  die("XML file not found.");
}
$xml = simplexml_load_file($file);
$stmt = $conn->prepare(
  "INSERT INTO students
  (roll_no,name,email,course,age)
  VALUES(?,?,?,?,?)"
);
$count = 0;
foreach ($xml->student as $student) {
  $roll_no =
    (string) $student->roll_no;
  $name =
    (string) $student->name;
  $email =
```

```
        (string) $student->email;
    $course =
        (string) $student->course;
    $age =
        (int) $student->age;
    $stmt->bind_param(
        "ssssi",
        $roll_no,
        $name,
        $email,
        $course,
        $age
    );
    if ($stmt->execute()) {
        $count++;
    }
}
echo "XML data imported successfully.";
echo "<br>";
echo "Total records: " . $count;
?>
```

Output :-

XML data imported successfully.
Total records: 3

3) Read data from students table of MySQL and store them in students.xml file.

Xml_export.php

```
<?php
require "config/db.php";
$result =
   $conn->query(
      "SELECT
         roll_no,
         name,
         email,
         course,
         age
       FROM students"
   );
$xml =
   new SimpleXMLElement(
      '<?xml version="1.0" encoding="UTF-8"?>
      <students/>'
   );
while (
   $student =
   $result->fetch_assoc()
) {

   $node =
      $xml->addChild("student");

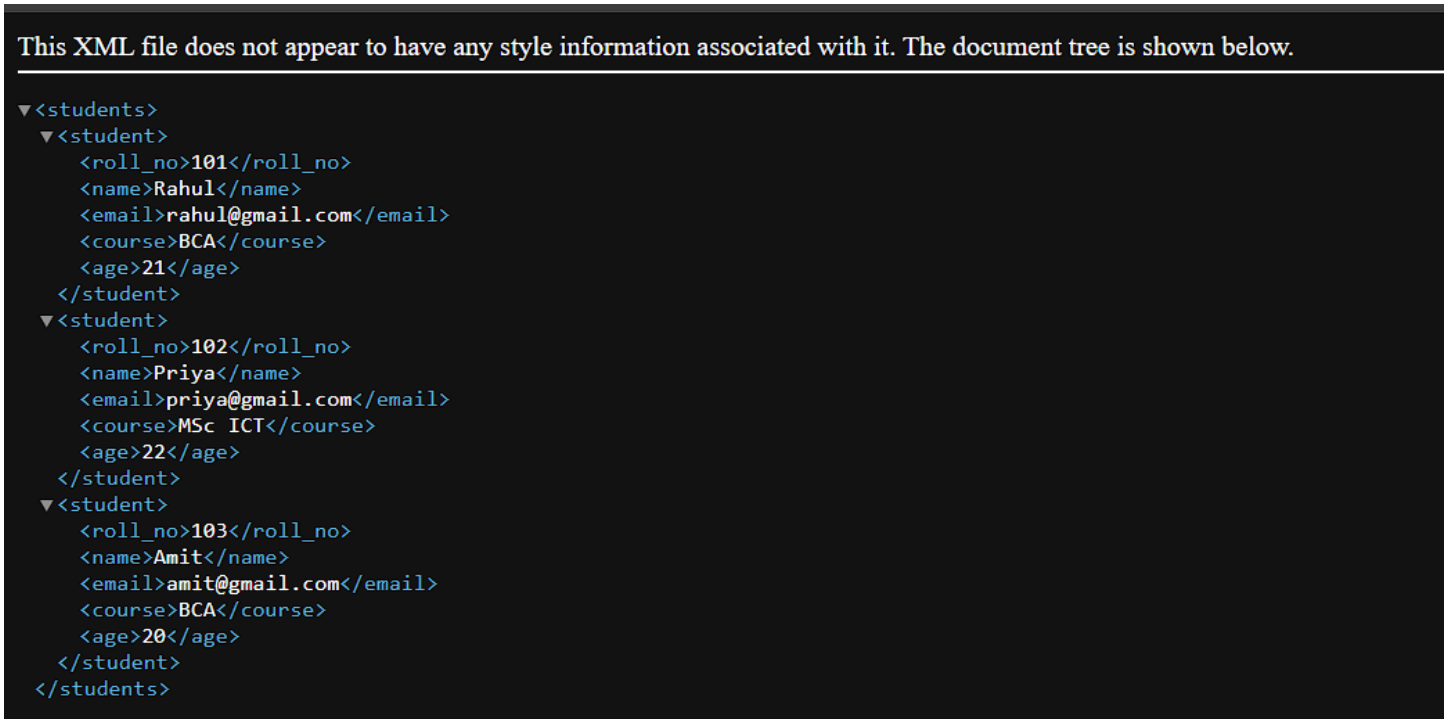
   $node->addChild(
      "roll_no",
      htmlspecialchars($student['roll_no'])
   );

   $node->addChild(
      "name",
      htmlspecialchars($student['name'])
   );

   $node->addChild(
      "email",
      htmlspecialchars($student['email'])
   );
```

```
    $node->addChild(
        "course",
        htmlspecialchars($student['course'])
    );
    $node->addChild(
        "age",
        $student['age']
    );
}
$file =
    "xml/students_export.xml";
$xml->asXML($file);
echo "MySQL data exported successfully.";
echo "<br>";
echo "<a href='$file'>Open XML File</a>";
?>
```

Output:



4) List files in a directory.

List_file.php

```
<?php
$directory = "uploads/products";
if (!is_dir($directory)) {
   die("Directory does not exist.");
}
$files =
   scandir($directory);
echo "<h2>Files</h2>";
echo "<ul>";
foreach ($files as $file) {
   if (
      $file == "." ||
      $file == ".."
   ) {
      continue;
   }
   $path =
      $directory .
      "/" .
      $file;
   if (is_file($path)) {
      echo "<li>";
      echo "<a href='$path'>";
      echo htmlspecialchars($file);
      echo "</a>";
      echo "</li>";
   }
}
echo "</ul>";
?>
```

Output:

Files

- product_6ad619f63d340.jpg

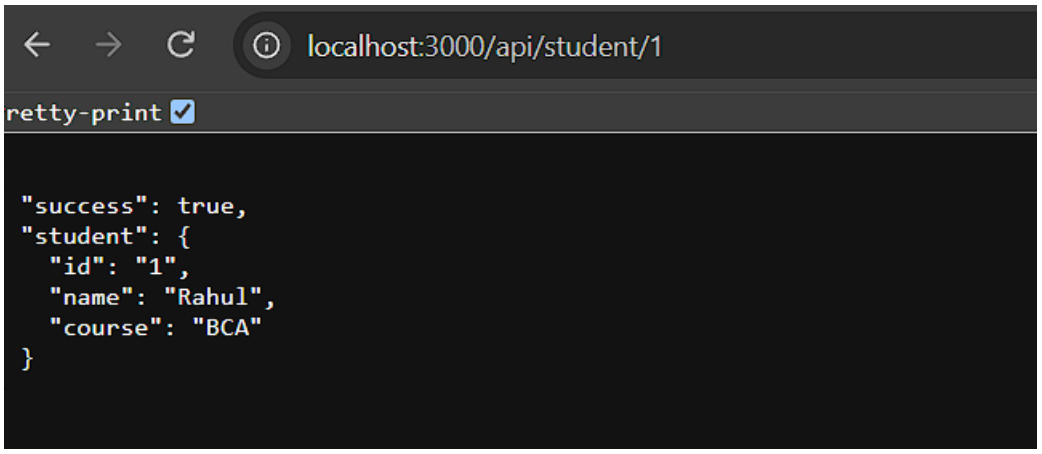
5) Call express rest api in php.

Express_server.js

```
import express from "express";
const app = express();
app.use(express.json());
app.get("/api/products", (req, res) => {
   const products = [
      {
         id: 1,
         pname: "T-Shirt",
         price: 499
      },
      {
         id: 2,
         pname: "Laptop",
         price: 55000
      }
   ];

   res.json({
      success: true,
      products: products
   });
});
app.get("/api/student/:id", (req, res) => {
   res.json({
      success: true,
      student: {
         id: req.params.id,
         name: "Rahul",
         course: "BCA"
      }
   });
});
app.listen(3000, () => {
   console.log(
      "Express server running on port 3000"
   );
});
```

Output:



6) Call php rest api in express.

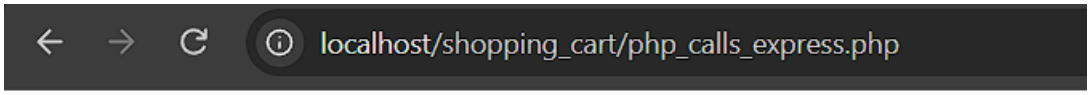
```
<?php
$url =
   "http://localhost:3000/api/products";
$ch =
   curl_init($url);
curl_setopt(
   $ch,
   CURLOPT_RETURNTRANSFER,
   true
);

curl_setopt(
   $ch,
   CURLOPT_HTTPHEADER,
   [
      "Accept: application/json"
   ]
);
$response =
   curl_exec($ch);
if ($response === false) {
   die(
      "cURL Error: " .
      curl_error($ch)
   );
}
```

```
curl_close($ch);

$data =
   json_decode(
      $response,
      true
   );
echo "<h2>Express API Response</h2>";
echo "<pre>";
print_r($data);
echo "</pre>";
?>
```

Output:



Express API Response

```
Array
(
    [success] => 1
    [products] => Array
        (
            [0] => Array
                (
                    [id] => 1
                    [pname] => T-Shirt
                    [price] => 499
                )

            [1] => Array
                (
                    [id] => 2
                    [pname] => Laptop
                    [price] => 55000
                )

        )

)
```